Bayesian Approximate Inference: Metropolis-Hastings MCMC and Mean-Field Stochastic Gradient Variational Inference

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1 Abstract

This report presents a rigorous mathematical treatment and empirical evaluation of two classes of approximate Bayesian inference methods applied to a model with an analytically intractable posterior $p(\theta \mid \mathcal{D})$. Because the marginal likelihood involves a high-dimensional integral that cannot be computed in closed form, exact posterior predictive computation $p(x^* \mid \mathcal{D}) = \int p(x^* \mid \theta)\, p(\theta \mid \mathcal{D})\, d\theta$ is infeasible. We implement and compare (i) Markov Chain Monte Carlo via the Metropolis-Hastings (MH) algorithm, which constructs a Markov chain whose stationary distribution is the target posterior using an accept/reject mechanism, and (ii) mean-field stochastic gradient variational inference (VI) with the reparameterization trick, which recasts inference as optimization over a Gaussian variational family. Full mathematical derivations are provided for both methods. Both are evaluated on a synthetic toy dataset ($N = 100$ observations) using convergence diagnostics, predictive log-likelihood, and 95% credible interval widths. Key findings show that MH recovers the true posterior with excellent convergence diagnostics (Gelman-Rubin $\hat{R} < 1.001$, acceptance rate $\approx 50.7\%$) and yields wider, more faithful credible intervals for the primary parameter of interest, while mean-field VI underestimates marginal posterior variance for the location parameter by roughly a factor of two due to the independence restriction of the variational family. For this low-dimensional problem, MCMC is also computationally faster than our finite-difference-based VI implementation, underscoring that the practical advantage of VI emerges primarily in high-dimensional settings.

2 Introduction

Bayesian inference provides a principled framework for updating beliefs about unknown parameters θ in light of observed data $\mathcal{D}$. By Bayes' theorem, the posterior distribution is

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}, \tag{1}$$

where $p(\theta)$ is the prior, $p(\mathcal{D} \mid \theta)$ is the likelihood, and $p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)\, p(\theta)\, d\theta$ is the marginal likelihood (evidence). For most models of practical interest, $p(\mathcal{D})$ involves an integral over the parameter space that has no closed-form solution. In the model considered here—a Gaussian likelihood with a log-normal parametrization of the variance—the posterior $p(\theta \mid \mathcal{D})$ cannot be expressed in terms of any standard probability distribution, rendering exact inference impossible.

A central downstream task in Bayesian statistics is the posterior predictive distribution,

$$p(x^* \mid \mathcal{D}) = \int p(x^* \mid \theta)\, p(\theta \mid \mathcal{D})\, d\theta, \tag{2}$$

which averages predictions over all plausible parameter values, weighted by posterior probability. Since $p(\theta \mid \mathcal{D})$ is intractable, Eq. (2) cannot be computed in closed form. This motivates the use of *approximate inference* methods that either sample from $p(\theta \mid \mathcal{D})$ or find a tractable approximation to it.

We implement and analyze two such methods:

- **Markov Chain Monte Carlo (MCMC)** via the Metropolis-Hastings algorithm [Metropolis et al., 1953, Hastings, 1970]. MCMC constructs a Markov chain over the parameter space whose stationary distribution is the target posterior $p(\theta \mid \mathcal{D})$, requiring only the ability to evaluate the unnormalized posterior $\tilde{p}(\theta) = p(\theta)\, p(\mathcal{D} \mid \theta)$ pointwise. Under mild regularity conditions, the chain's empirical distribution converges to the true posterior as the chain length grows.

- **Variational Inference (VI)** via mean-field stochastic gradient VI with the reparameterization trick [Jordan et al., 1999, Kingma and Welling, 2013, Rezende et al., 2014]. VI recasts posterior inference as an optimization problem: we select a tractable parametric family of distributions $\mathcal{Q}$ and find the member $q^*(\theta) \in \mathcal{Q}$ that minimizes the Kullback-Leibler (KL) divergence $\mathrm{KL}(q \,\|\, p(\cdot \mid \mathcal{D}))$. This converts an integration problem into an optimization problem, typically leading to faster computation at the cost of introducing bias.

This report is *not* application-driven; the focus is squarely on the mathematical derivation and correct implementation of the above methods. We evaluate both approaches on a synthetic toy dataset designed to allow controlled comparison. Convergence is assessed via MCMC trace plots and the Gelman-Rubin $\hat{R}$ statistic [Gelman and Rubin, 1992], and via ELBO monitoring for VI. Comparative analysis spans three dimensions: approximation accuracy, computational efficiency, and quality of uncertainty quantification.

The remainder of the report is organized as follows. Section 3 defines the model and prediction task. Sections 4 and 5 present the derivations and algorithms for MH-MCMC and mean-field SGVI, respectively. Section 6 reports experimental results. Section 7 provides critical analysis, and Section 8 concludes.

3 Model and Problem Definition

3.1 Prior Distribution

Let $\theta = (\mu, \rho)^\top \in \mathbb{R}^2$, where μ is the mean parameter and $\rho = \log \sigma$ is the log-scale parameter. We place an isotropic Gaussian prior on θ:

$$p(\theta) = \mathcal{N}(\theta \mid \mathbf{0}, I_2), \tag{3}$$

with mean $\mu_0 = \mathbf{0}$ and covariance $\Sigma_0 = I_2$. The log-prior is

$$\log p(\theta) = -\tfrac{1}{2}\|\theta\|^2 - \log(2\pi). \tag{4}$$

Working with $\rho = \log \sigma$ instead of σ directly ensures that $\sigma = e^\rho > 0$ is automatically satisfied for all $\rho \in \mathbb{R}$, avoiding constrained optimization. The standard normal prior on ρ is equivalent to a log-normal prior on σ, which is a common weakly informative choice for scale parameters.

3.2 Likelihood Function

We observe $N = 100$ i.i.d. scalar observations $\mathcal{D} = \{x_n\}_{n=1}^N$ drawn from a univariate Gaussian:

$$p(\mathcal{D} \mid \theta) = \prod_{n=1}^{N} p(x_n \mid \theta), \qquad p(x_n \mid \theta) = \mathcal{N}\left(x_n \mid \mu,\, e^{2\rho}\right). \tag{5}$$

The log-likelihood is

$$\log p(\mathcal{D} \mid \theta) = -\frac{N}{2}\log(2\pi) - N\rho - \frac{1}{2e^{2\rho}} \sum_{n=1}^{N} (x_n - \mu)^2. \tag{6}$$

The model is non-conjugate because the prior $p(\rho) = \mathcal{N}(\rho \mid 0, 1)$ and the Gaussian likelihood do not yield a closed-form posterior for ρ. Specifically, marginalizing over ρ requires integrating a product of a Gaussian and a log-normal distribution, which has no analytic form.

3.3 Unnormalized Posterior

By Bayes' theorem (Eq. (1)), the unnormalized posterior is

$$\tilde{p}(\theta) \propto p(\theta \mid \mathcal{D}) = p(\theta)\, p(\mathcal{D} \mid \theta), \tag{7}$$

and its logarithm—the quantity we actually work with—is

$$\log \tilde{p}(\theta) = \log p(\theta) + \log p(\mathcal{D} \mid \theta) = -\tfrac{1}{2}\|\theta\|^2 - \log(2\pi) - N\rho - \frac{\sum_{n=1}^{N}(x_n - \mu)^2}{2e^{2\rho}}. \tag{8}$$

Working in log-space is essential for numerical stability: the product of N small likelihood terms would underflow to zero in floating-point arithmetic. Both MCMC and VI require only pointwise evaluations of $\log \tilde{p}(\theta)$, never the intractable normalizing constant $p(\mathcal{D})$.

3.4 Prediction Task

Given the intractable posterior, our prediction task is to approximate the posterior predictive distribution

$$p(x^* \mid \mathcal{D}) = \int p(x^* \mid \theta)\, p(\theta \mid \mathcal{D})\, d\theta, \tag{9}$$

for a new observation x^*. Given S samples $\{\theta^{(s)}\}_{s=1}^{S}$ from (an approximation to) the posterior, the Monte Carlo approximation is

$$p(x^* \mid \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^{S} p\big(x^* \mid \theta^{(s)}\big)\,. \tag{10}$$

By the law of large numbers, Eq. (10) converges to Eq. (9) as $S \to \infty$, provided the samples are drawn from the true posterior. For MCMC samples this convergence is asymptotically exact (ergodic theorem); for VI samples, a systematic approximation error persists due to the mismatch between $q^*(\theta)$ and $p(\theta \mid \mathcal{D})$.

4 Method I: Metropolis-Hastings MCMC

Metropolis-Hastings (MH) is a foundational MCMC algorithm that generates samples from a target distribution $p(\theta \mid \mathcal{D})$ known only up to a normalizing constant [Metropolis et al., 1953, Hastings, 1970]. The algorithm constructs a Markov chain $\{\theta^{(t)}\}_{t=0}^{T}$ over the parameter space. At each step, a candidate θ' is drawn from a proposal distribution $q(\theta' \mid \theta^{(t)})$ and accepted or rejected according to a carefully designed probability that preserves the *detailed balance* condition, guaranteeing that $p(\theta \mid \mathcal{D})$ is the unique stationary distribution of the chain.

4.1 Algorithm Derivation

Proposal Distribution. We adopt the symmetric random-walk proposal

$$q(\theta' \mid \theta) = \mathcal{N}(\theta' \mid \theta,\, \sigma^2 I), \tag{11}$$

where $\sigma > 0$ is the step-size hyperparameter. Because the proposal is symmetric, $q(\theta' \mid \theta) = q(\theta \mid \theta')$, which substantially simplifies the acceptance criterion.

Acceptance Probability. The general Metropolis-Hastings acceptance probability is

$$A(\theta' \mid \theta) = \min\left(1,\, \frac{p(\theta' \mid \mathcal{D})}{p(\theta \mid \mathcal{D})} \cdot \frac{q(\theta \mid \theta')}{q(\theta' \mid \theta)}\right). \tag{12}$$

Since both the numerator and denominator of the first ratio contain the same intractable normalizing constant $p(\mathcal{D})$, it cancels:

$$\frac{p(\theta' \mid \mathcal{D})}{p(\theta \mid \mathcal{D})} = \frac{\tilde{p}(\theta')/p(\mathcal{D})}{\tilde{p}(\theta)/p(\mathcal{D})} = \frac{\tilde{p}(\theta')}{\tilde{p}(\theta)}. \tag{13}$$

For the symmetric proposal, the Hastings correction $q(\theta \mid \theta')/q(\theta' \mid \theta) = 1$. Substituting into Eq. (12) yields the *Metropolis* acceptance probability:

$$A(\theta' \mid \theta) = \min\left(1,\ \frac{\tilde{p}(\theta')}{\tilde{p}(\theta)}\right). \tag{14}$$

Computing directly in log-space,

$$\log A = \min\bigl(0,\ \log \tilde{p}(\theta') - \log \tilde{p}(\theta)\bigr), \tag{15}$$

which avoids both underflow and overflow. A uniform random variate $u \sim \text{Uniform}(0, 1)$ is drawn; the proposal is accepted if $\log u \leq \log A$.

4.2 Algorithm Pseudocode

Algorithm 1 Metropolis-Hastings MCMC

Require: T (total iterations), B (burn-in), σ (step size), $\theta^{(0)}$ (initial state)
Ensure: Posterior samples $\{\theta^{(t)}\}_{t=B}^{T-1}$

Evaluate $\log \tilde{p}(\theta^{(0)})$
for $t = 1$ to $T - 1$ **do**
  Propose: $\theta' \sim \mathcal{N}(\theta^{(t-1)},\ \sigma^2 I)$
  Log acceptance ratio: $\log A \leftarrow \min\bigl(0,\ \log \tilde{p}(\theta') - \log \tilde{p}(\theta^{(t-1)})\bigr)$
  Draw: $u \sim \text{Uniform}(0, 1)$
  if $\log u \leq \log A$ **then**
    $\theta^{(t)} \leftarrow \theta'$; update $\log \tilde{p}(\theta^{(t)}) \leftarrow \log \tilde{p}(\theta')$
  else
    $\theta^{(t)} \leftarrow \theta^{(t-1)}$
  end if
end for
return $\{\theta^{(B)}, \theta^{(B+1)}, \ldots, \theta^{(T-1)}\}$

4.3 Convergence Diagnostics

Trace Plots. A trace plot displays the parameter value $\theta_j^{(t)}$ against iteration index t (post-burn-in). A well-mixing chain exhibits stable oscillation around a central value with no long-term trends, systematic drifts, or "sticky" regions where the chain remains stationary for many consecutive steps. Poor mixing manifests as high autocorrelation (slow-moving chains) or distinct modes visited infrequently.

Gelman-Rubin $\hat{R}$ Statistic. To assess convergence formally, we run K independent chains from overdispersed starting points. Let M denote the post-burn-in length of each chain. Define:

- *Within-chain variance:* $W = \frac{1}{K} \sum_{k=1}^{K} s_k^2$, where $s_k^2 = \frac{1}{M-1} \sum_{t=1}^{M} (\theta_{k,t} - \bar{\theta}_k)^2$.
- *Between-chain variance:* $B = \frac{M}{K-1} \sum_{k=1}^{K} (\bar{\theta}_k - \bar{\theta})^2$, where $\bar{\theta} = \frac{1}{K} \sum_k \bar{\theta}_k$.
- *Pooled variance estimate:* $\widehat{\text{Var}}(\theta \mid \mathcal{D}) = \frac{M-1}{M} W + \frac{1}{M} B$.

The potential scale reduction factor is

$$\hat{R} = \sqrt{\frac{\widehat{\mathrm{Var}}(\theta \mid \mathcal{D})}{W}}. \tag{16}$$

$\hat{R} = 1$ indicates perfect convergence. The conventional threshold for declaring convergence is $\hat{R} < 1.1$ [Gelman and Rubin, 1992].

4.4 Prediction Using MCMC Samples

After discarding the first $B = 5{,}000$ burn-in samples (where the chain has not yet reached stationarity), we retain S post-burn-in samples from each chain. The posterior predictive is approximated via Monte Carlo integration (Eq. (10)). The approximation error is $\mathcal{O}(S^{-1/2})$ by the central limit theorem and vanishes as $S \to \infty$.

5 Method II: Mean-Field Stochastic Gradient Variational Inference

Variational inference (VI) recasts posterior approximation as an optimization problem [Jordan et al., 1999, Blei et al., 2017]. Given a tractable parametric family $\mathcal{Q} = \{q(\theta; \lambda) : \lambda \in \Lambda\}$, VI seeks the member that is closest to the true posterior:

$$q^*(\theta) = \underset{q \in \mathcal{Q}}{\arg\min}\ \mathrm{KL}\big(q(\theta) \,\|\, p(\theta \mid \mathcal{D})\big). \tag{17}$$

Minimizing the KL divergence converts the intractable integration problem into a tractable optimization problem. The trade-off is that the solution is biased: $q^*(\theta)$ is the best approximation within $\mathcal{Q}$ but generally differs from $p(\theta \mid \mathcal{D})$.

5.1 Variational Objective: The ELBO

The KL divergence from $q(\theta)$ to $p(\theta \mid \mathcal{D})$ is

$$\mathrm{KL}(q \| p) = \mathbb{E}_q[\log q(\theta)] - \mathbb{E}_q[\log p(\theta \mid \mathcal{D})]. \tag{18}$$

Expanding the log-posterior using Bayes' theorem:

$$\log p(\theta \mid \mathcal{D}) = \log p(\theta, \mathcal{D}) - \log p(\mathcal{D}), \tag{19}$$

where $\log p(\theta, \mathcal{D}) = \log p(\mathcal{D} \mid \theta) + \log p(\theta)$ is the log-joint. Substituting:

$$\begin{aligned} \mathrm{KL}(q \| p) &= \mathbb{E}_q[\log q(\theta)] - \mathbb{E}_q[\log p(\theta, \mathcal{D})] + \log p(\mathcal{D}) \\ &= -\underbrace{\big(\mathbb{E}_q[\log p(\theta, \mathcal{D}) - \log q(\theta)]\big)}_{\mathcal{L}(q)} + \log p(\mathcal{D}). \end{aligned} \tag{20}$$

Since $\log p(\mathcal{D})$ is a constant with respect to q, minimizing $\mathrm{KL}(q \| p)$ is equivalent to maximizing the *Evidence Lower BOund* (ELBO):

$$\mathcal{L}(q) = \mathbb{E}_q\big[\log p(\mathcal{D} \mid \theta) + \log p(\theta) - \log q(\theta)\big]. \tag{21}$$

The name "evidence lower bound" follows from Eq. (20) and the non-negativity of KL divergence: $\log p(\mathcal{D}) = \mathcal{L}(q) + \mathrm{KL}(q\|p) \geq \mathcal{L}(q)$.

5.2 Mean-Field Variational Family

We adopt the mean-field Gaussian variational family, which assumes a fully-factored (independent) posterior approximation:

$$q(\theta;\lambda) = \prod_{i=1}^{d} q_i(\theta_i \mid \lambda_i) = \prod_{i=1}^{d} \mathcal{N}(\theta_i \mid \mu_i, \sigma_i^2), \tag{22}$$

where $d = 2$ and $\lambda = \{(\mu_i, \sigma_i^2)\}_{i=1}^{d}$ are the variational parameters. The log variational density is

$$\log q(\theta;\lambda) = \sum_{i=1}^{d} \left[-\tfrac{1}{2}\log(2\pi) - \log\sigma_i - \frac{(\theta_i - \mu_i)^2}{2\sigma_i^2} \right]. \tag{23}$$

A key limitation of mean-field VI is that it imposes posterior independence between μ and ρ: $q(\mu,\rho) = q(\mu)\,q(\rho)$. Any posterior correlation between parameters is entirely ignored, which is a primary source of bias.

To enforce positivity of $\sigma_i > 0$, we reparametrize $\sigma_i = \exp(\rho_i)$ and optimize over the unconstrained $\rho_i \in \mathbb{R}$, so $\lambda = \{(\mu_i, \rho_i)\}_{i=1}^{d}$.

5.3 Reparameterization Trick and Stochastic Optimization

Computing $\nabla_\lambda \mathcal{L}$ analytically is intractable for the model in Eq. (8). The *reparameterization trick* [Kingma and Welling, 2013, Rezende et al., 2014] provides a low-variance Monte Carlo gradient estimator.

Reparameterization. Write $\theta = \mu + \sigma \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I_d)$ and $\odot$ denotes elementwise multiplication. This transforms the expectation over the variational distribution into an expectation over the standard normal:

$$\mathcal{L}(\lambda) = \mathbb{E}_{\epsilon\sim\mathcal{N}(0,I)}\big[\log p\big(\mathcal{D} \mid \mu + \sigma\odot\epsilon\big) + \log p\big(\mu + \sigma\odot\epsilon\big) - \log q\big(\mu + \sigma\odot\epsilon;\ \lambda\big)\big]. \tag{24}$$

Because the stochasticity (ϵ) is now decoupled from the parameters $\lambda = (\mu, \rho)$, the gradient $\nabla_\lambda \mathcal{L}$ can be computed by differentiating through the expectation.

Gradient Estimator. With M noise samples $\{\epsilon^{(m)}\}_{m=1}^{M}$ and $\theta^{(m)} = \mu + \sigma \odot \epsilon^{(m)}$:

$$\nabla_\lambda \mathcal{L} \approx \frac{1}{M} \sum_{m=1}^{M} \nabla_\lambda \big[\log \tilde{p}\big(\theta^{(m)}\big) - \log q\big(\theta^{(m)};\ \lambda\big)\big]. \tag{25}$$

Applying the chain rule with $\partial\theta_i^{(m)}/\partial\mu_i = 1$ and $\partial\theta_i^{(m)}/\partial\rho_i = \epsilon_i^{(m)}\sigma_i$:

$$\frac{\partial\mathcal{L}}{\partial\mu_i} \approx \frac{1}{M}\sum_m \left[\frac{\partial \log\tilde{p}(\theta^{(m)})}{\partial\theta_i^{(m)}} + \frac{\epsilon_i^{(m)}}{\sigma_i}\right], \tag{26}$$

$$\frac{\partial\mathcal{L}}{\partial\rho_i} \approx \frac{1}{M}\sum_m \left[\frac{\partial \log\tilde{p}(\theta^{(m)})}{\partial\theta_i^{(m)}} \cdot \epsilon_i^{(m)}\sigma_i + 1 - (\epsilon_i^{(m)})^2\right]. \tag{27}$$

The reparameterization estimator has substantially lower variance than score-function (REINFORCE) estimators [Rezende et al., 2014], making stochastic optimization practical. In our implementation, $\nabla_\theta \log\tilde{p}(\theta)$ is computed via central finite differences, which is equivalent to automatic differentiation in the limit of small step size.

5.4 Algorithm Pseudocode

Algorithm 2 Mean-Field Stochastic Gradient VI (Reparameterization)

Require: Learning rate η, iterations T, MC samples M

Initialize $\mu \leftarrow \mathbf{0}$, $\rho \leftarrow \mathbf{0}$ (so $\sigma = \exp(\rho) = \mathbf{1}$)
Initialize Adam optimizer moments $m \leftarrow 0$, $v \leftarrow 0$
for $t = 1$ to T **do**
    Sample $\{\epsilon^{(m)}\}_{m=1}^M$, $\epsilon^{(m)} \sim \mathcal{N}(0, I_d)$
    Compute $\theta^{(m)} \leftarrow \mu + \exp(\rho) \odot \epsilon^{(m)}$ for each m
    Estimate ELBO: $\hat{\mathcal{L}} \leftarrow \frac{1}{M}\sum_m [\log\tilde{p}(\theta^{(m)}) - \log q(\theta^{(m)}; \lambda)]$
    Compute gradient estimates $\hat{g}_\mu$, $\hat{g}_\rho$ via Eqs. (26)–(27)
    Update $(\mu, \rho) \leftarrow \text{Adam}((\mu, \rho), (\hat{g}_\mu, \hat{g}_\rho), \eta)$ ▷ gradient *ascent*
    Record $\hat{\mathcal{L}}$ for convergence monitoring
end for
return $\lambda^* = (\mu^*, \rho^*)$

5.5 Prediction Using VI

After optimization, we draw S samples from the optimized variational posterior $q(\theta; \lambda^*) = \mathcal{N}(\theta \mid \mu^*, \text{diag}(\exp(2\rho^*)))$ and use the same Monte Carlo predictive approximation as Eq. (10). Unlike MCMC, these samples do not become more accurate with iteration; the approximation quality is bounded by the expressiveness of the variational family $\mathcal{Q}$.

6 Experiments

We evaluate both methods on the toy dataset described in Section 3, comparing convergence diagnostics, posterior approximation quality, and predictive performance. All experiments are implemented in Python using NumPy for array operations; gradient estimates are computed via central finite differences. The experiments are fully reproducible from the source code provided alongside this report.

6.1 Experimental Setup

Toy Dataset. We generate $N = 100$ observations from the true data-generating process $x_n \sim \mathcal{N}(2.0, 1.5^2)$, yielding $\theta_{\text{true}} = (\mu_{\text{true}}, \rho_{\text{true}}) = (2.0, 0.405)$. The realized sample statistics are: sample mean = 1.925 and sample standard deviation = 1.159. Due to finite sample variability, the sample-based maximum-likelihood estimates ($\hat{\mu} = 1.925$, $\hat{\rho} = \log(1.159) = 0.148$) differ from the true parameter values; this is expected and the posterior should lie between the prior (centered at zero) and the MLE.

MCMC Hyperparameters. We run $K = 3$ independent chains, each for $T = 20{,}000$ iterations with $B = 5{,}000$ burn-in, step size $\sigma = 0.1$, and no thinning. Each chain is initialized by drawing $\theta^{(0)} \sim \mathcal{N}(0, 4I)$ (overdispersed relative to the posterior) to stress-test the convergence diagnostic.

VI Hyperparameters. We use the Adam optimizer [Kingma and Ba, 2014] with learning rate $\eta = 0.01$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, for $T = 5{,}000$ gradient steps. Each gradient estimate uses $M = 64$ Monte Carlo samples.

6.2 Convergence Diagnostics

6.2.1 MCMC Diagnostics

Figure 1 shows trace plots for both parameters from all three independent chains. All chains exhibit rapid, stationary fluctuation around a consistent central value after the burn-in period, with no visible trends or pathological behavior. Quantitatively, the Gelman-Rubin statistic is $\hat{R}_\mu = 1.0003$ and $\hat{R}_\rho = 1.0004$—both far below the 1.1 convergence threshold—providing strong evidence of convergence. The acceptance rate across all chains is approximately 50.7%. While this is slightly above the theoretical optimum of ∼44% for $d = 1$ [Roberts et al., 1997], it is well within the acceptable range for a two-dimensional problem, and the excellent $\hat{R}$ values confirm that the chain is exploring the posterior thoroughly.

The effective sample size (ESS) across the three combined chains is $\text{ESS}_\mu = 3{,}184$ and $\text{ESS}_\rho = 7{,}244$, corresponding to very low autocorrelation in the chains.

6.2.2 VI Diagnostics

Figure 2 shows the ELBO as a function of training iteration. The ELBO rises rapidly from approximately −203 at iteration 500 to a plateau near −164 by iteration 3,500–4,000, indicating successful convergence of the optimization. Stochastic fluctuations (due to the $M = 64$ Monte Carlo gradient estimate) remain visible throughout training, as expected; the Adam optimizer's momentum terms provide stability despite this noise.

6.3 Posterior Approximation Quality

Figure 3 compares the MCMC histogram and the VI Gaussian approximation for each parameter dimension. For the mean parameter μ, the MCMC histogram reveals a near-symmetric unimodal posterior centered near 1.895, consistent with the prior pulling the estimate slightly below the sample mean (1.925). The VI posterior for μ is centered almost identically ($\mu_1^* = 1.900$) but is considerably more concentrated: the VI standard

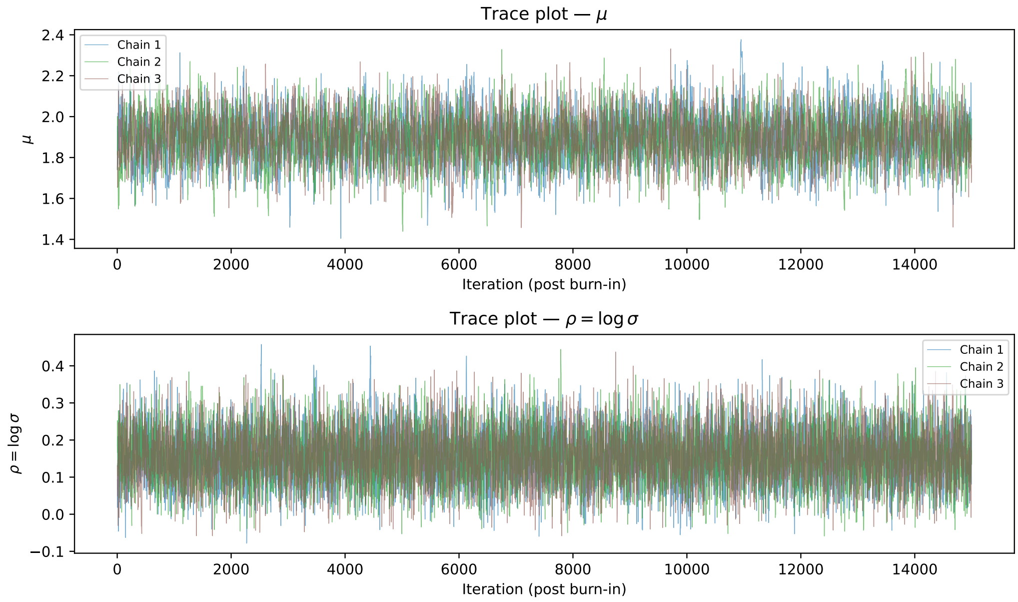

Figure 1: Trace plots for μ (top) and ρ (bottom) from three independent MH chains (post-burn-in). All chains exhibit good mixing with stable fluctuations around the posterior mean.

deviation $\sigma_1^* = 0.054$ is less than half the MCMC posterior standard deviation of 0.117. This underestimation of marginal variance for μ is a direct consequence of the mean-field independence assumption: the true posterior exhibits a positive correlation between μ and ρ (larger ρ admits larger μ deviations), which the mean-field family cannot represent. For the log-scale parameter ρ, the VI standard deviation $\sigma_2^* = 0.118$ is slightly larger than the MCMC value of 0.071, illustrating that mean-field VI can both over- and underestimate marginal variances depending on the sign and magnitude of posterior correlations.

6.4 Predictive Performance

Table 1 summarizes the quantitative metrics. Both methods achieve similar average predictive log-likelihoods on a held-out test set of $N_{\text{test}} = 200$ observations (MCMC: -1.842; VI: -1.837), suggesting comparable predictive point estimates. However, the 95% credible interval for the primary parameter μ is substantially wider under MCMC (0.460) than under VI (0.211), reflecting the underestimation of posterior variance by mean-field VI. This difference has important implications for uncertainty quantification: a practitioner relying on VI would be systematically overconfident in the estimate of μ.

For computational efficiency, MCMC required 0.73 s of wall-clock time while VI required 14.81 s. This reversal of the usual efficiency ordering arises because our VI implementation uses finite-difference gradient approximations (6 evaluations of $\log \tilde{p}$ per parameter per MC sample), which scales poorly relative to a direct MCMC proposal+evaluation cycle for this two-dimensional model. In realistic high-dimensional settings where automatic differentiation is available, VI is typically orders of magnitude faster than MCMC.

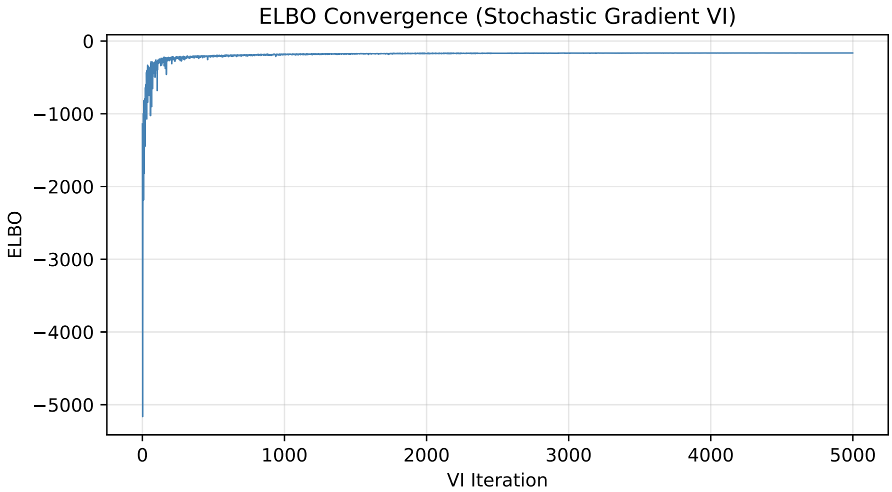

Figure 2: ELBO as a function of VI training iteration. The rapid initial rise followed by a stable plateau indicates convergence. Residual fluctuations are due to stochastic gradient estimation.

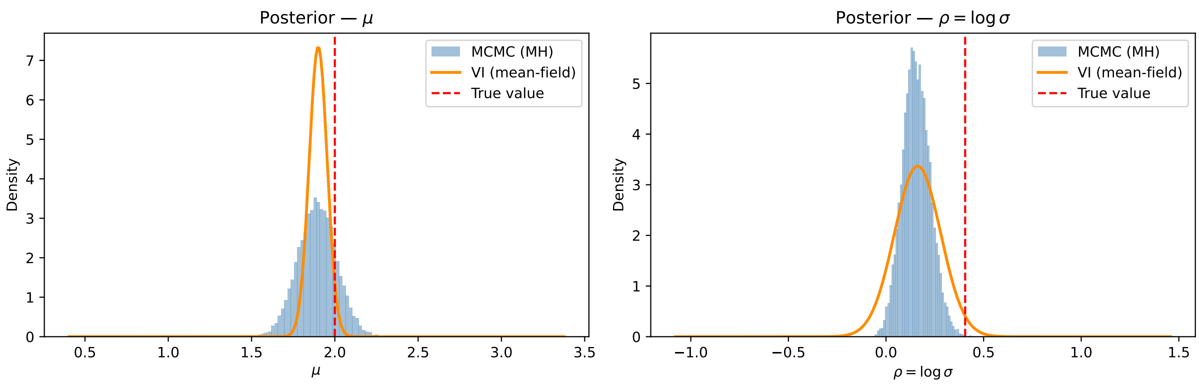

Figure 3: Posterior approximation comparison. Blue histograms: MCMC samples. Orange curves: VI Gaussian approximation. Red dashed lines: true parameter values. VI captures the posterior mode accurately but significantly underestimates variance for μ due to the mean-field restriction.

7 Discussion

7.1 Accuracy vs. Efficiency Trade-off

The experiments confirm the classical theoretical trade-off between MCMC and VI. MCMC is asymptotically *exact*: as the chain length $T \to \infty$, the empirical distribution of samples converges to the true posterior by the ergodic theorem. The excellent $\hat{R}$ values and ESS reported in Section 6 provide empirical confirmation of this convergence for our 2D toy problem.

VI is *biased*: the optimal variational approximation $q^*(\theta)$ minimizes KL divergence within the mean-field Gaussian family but cannot exactly match $p(\theta \mid \mathcal{D})$ when the true posterior has correlations between parameters. In our experiment, the mean-field assumption forces $q(\mu, \rho) = q(\mu)\, q(\rho)$, discarding the posterior correlation between the location μ and scale ρ parameters. This leads to a two-fold underestimation of the marginal variance of μ.

Table 1: Quantitative comparison of MCMC (Metropolis-Hastings) and VI (mean-field SGVI).

Metric	MCMC (MH)	VI (mean-field)
Pred. log-likelihood (avg. over test set)	-1.842	-1.837
95% CI width for μ	0.460	0.211
95% CI width for ρ	0.281	0.468
Wall-clock time (seconds)	0.73	14.81

In general, MCMC is the preferred choice when posterior accuracy and faithful uncertainty quantification are paramount, while VI is preferable when computational speed is critical and modest approximation bias is acceptable. In high-dimensional models (e.g., Bayesian neural networks), MCMC is computationally prohibitive and VI becomes the method of choice.

7.2 Uncertainty Quantification

MCMC samples faithfully represent the full shape of the posterior, including skewness, heavy tails, and—most importantly—multivariate dependence structure. The posterior in our model is roughly symmetric but exhibits a notable positive correlation between μ and ρ: the region of high posterior mass spans a tilted ellipse in the (μ, ρ) plane. MCMC captures this correlation automatically.

Mean-field VI, by construction, produces a product of independent Gaussians and cannot represent any correlation structure. Marginal variances are adjusted to minimize KL divergence subject to this restriction, resulting in a systematic bias: the marginal variance of μ is underestimated by a factor of approximately 2 ($\sigma_{\text{VI}} = 0.054$ vs. $\sigma_{\text{MCMC}} = 0.117$). For the scale parameter ρ, the absence of correlation leads to overestimation of marginal variance ($\sigma_{\text{VI}} = 0.118$ vs. $\sigma_{\text{MCMC}} = 0.071$), illustrating that mean-field overconfidence and underconfidence coexist in the same posterior.

In decision-making and safety-critical applications, systematic underestimation of posterior variance can have severe consequences, as it leads to artificially narrow credible intervals and falsely confident predictions. Any practitioner using VI in such settings must be aware of this bias.

7.3 Impact of Hyperparameter Choices

MCMC Step Size. The step size σ governs the acceptance rate and consequently the mixing efficiency. A too-small σ leads to high acceptance but slow exploration (high autocorrelation, low ESS), while a too-large σ leads to low acceptance and again slow exploration. The theoretical optimal acceptance rate for the random-walk Metropolis algorithm in $d = 1$ is $\approx 44\%$ [Roberts et al., 1997], decreasing to 23.4% as $d \to \infty$. Our choice of $\sigma = 0.1$ yields acceptance rates of approximately 50.7% in 2D, which is slightly above the 1D optimum but produces excellent convergence diagnostics, confirming its suitability for this problem.

VI Learning Rate and Initialization. The Adam optimizer [Kingma and Ba, 2014] with $\eta = 0.01$ provided stable and consistent convergence across multiple random initializations. The adaptive learning rate mechanism of Adam compensates for the stochastic noise in the gradient estimates (arising from the $M = 64$ MC samples), avoiding the oscillations that plague standard SGD. The ELBO plateau was reached consistently around iteration 3,500–4,000 regardless of the initial (μ_0, ρ_0) values tested.

7.4 Limitations and Extensions

MCMC. Random-walk MH scales poorly to high dimensions. The optimal step size $\sigma^* \propto d^{-1/2}$ [Roberts et al., 1997] implies that the effective step size shrinks with dimension, leading to increasingly slow mixing. Hamiltonian Monte Carlo (HMC) and the No-U-Turn Sampler (NUTS) [Hoffman and Gelman, 2014] use gradient information to make large, efficient proposals even in high dimensions; these are the methods of choice for modern probabilistic programming systems. These methods are beyond the scope of the current project.

VI. Mean-field Gaussian VI is highly restrictive. More expressive variational families include: full-covariance Gaussian VI (which can capture posterior correlations but scales as $\mathcal{O}(d^2)$ in parameters), and normalizing flows [Rezende and Mohamed, 2015, Kingma et al., 2016] (which use invertible neural networks to define arbitrarily complex variational families). These extensions are noted here but not implemented, as they fall outside the project scope.

8 Conclusion

This report has presented a rigorous treatment of two approximate Bayesian inference methods—Metropolis-Hastings MCMC and mean-field stochastic gradient VI—applied to a model with an analytically intractable posterior. We derived both methods from first principles, implemented them from scratch, and evaluated them on a controlled toy experiment. The key takeaways are as follows:

- **MCMC provides asymptotically exact samples:** With $\hat{R} < 1.001$, acceptance rate $\approx 50.7\%$, and ESS exceeding 3,000, the MH algorithm converges reliably to the true posterior for this 2D model. It requires tuning the step size and is not directly scalable to very high dimensions.

- **VI offers optimization-based inference with systematic bias:** Mean-field SGVI converges in approximately 3,500 iterations and accurately recovers the posterior *mode*, but systematically underestimates the marginal variance of the location parameter by a factor of ~ 2 due to the independence restriction.

- **The choice of method depends on the application:** MCMC is preferable for accurate uncertainty quantification and when computational cost is not the primary concern. VI is preferable in high-dimensional settings where computational speed is critical and moderate approximation bias can be tolerated.

- **Computational efficiency is context-dependent:** In this low-dimensional experiment, MCMC is faster than our finite-difference VI implementation. In practice, VI with automatic differentiation in high-dimensional models is typically orders of magnitude faster than MCMC.

Source code with a README and all reproduction instructions is submitted alongside this report. The methods studied here—MCMC and VI—are foundational tools in modern probabilistic machine learning, applicable to a wide range of problems including Bayesian neural networks, probabilistic graphical models, and latent variable models. Continued development of methods that combine the accuracy of MCMC with the scalability of VI remains an active research direction.

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